

AYMANE AIT DADS

AI Algorithm Research Intern - Neuromorphic Computing | LLM Fine-Tuning · Sparse Neural Architectures · Ablation Studies

Ayman.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Achieved top 10% public leaderboard (0.80/1.0) in the NVIDIA Nemotron Reasoning Kaggle Competition (\$106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters), conducting 20+ single-variable ablation studies to isolate optimal training configurations for event-driven, sparse reasoning architectures. Ranked 1st in EURECOM class (0.791 AUC) on an AI-generated image detection benchmark and submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark, demonstrating reproducible research-grade methodology applied to novel neural architectures with 428M-parameter transformer backbones. Core competencies span neuro-inspired algorithm design, PyTorch-based model training, CUDA kernel debugging, sparse MoE routing, chain-of-thought reasoning, and massively parallel GPU experimentation on Blackwell hardware. Available for internship in South America with no work authorization restrictions; studies based at EURECOM, Sophia Antipolis, France.

CORE COMPETENCIES

Neuromorphic Algorithms | Sparse Neural Architectures | Event-Driven Processing | Ablation Studies | LLM Fine-Tuning | Neuro-Inspired Computing | Massively Parallel Processing | Algorithm Research

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a neuro-inspired clustering pipeline combining Random Forest and K-Means on **100K+ fiber-optic network records**, mapping customer performance profiles to 5 commercial service tiers used directly by the network optimization team.
- Applied feature engineering and ablation-style model evaluation across upload speed, latency, and jitter signals, **reducing manual analysis time by ~60%** through end-to-end automation of the classification workflow.
- Developed a scalable unsupervised segmentation algorithm classifying network quality across **thousands of nodes**, producing maintenance recommendations validated against real operational thresholds.
- Delivered stakeholder presentations translating **algorithmic model outputs** into actionable network maintenance strategies, bridging research findings and engineering decision-making for a cross-functional team.

PROJECTS

NVIDIA Nemotron Reasoning Challenge - Kaggle

Kaggle · In Progress (June 2026) · Top 10% Public Leaderboard · \$106K+ Prize Pool

Competition

- Achieved **top 10% public leaderboard (0.80/1.0)** fine-tuning a 30B Mamba-Transformer MoE (3.5B active via sparse MoE routing, 52 layers) using LoRA (r=32, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples across 6 logic puzzle types evaluated by exact match.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order - including monkey-patched CUDA allocators, stubbed Mamba/Cutlass imports, and pure PyTorch RMSNorm fallbacks - and conducted 20+ single-variable ablations confirming completion-only loss (+0.01 AUC) and that sequence packing catastrophically degraded score from 0.80 to 0.64.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · TRL · Triton · Mamba

AI-Generated Image Detection - NTIRE 2026 @ EURECOM Course + NTIRE 2026 @ CVPR CodaBench · 2025 · Ranked 1st in Class

CVPR & EURECOM ImSecu

- Ranked **1st in EURECOM class private leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M parameters, pretrained on 400M image-text pairs) with a custom MLP head and dual learning rates, training on 250K images spanning 25+ generator types with FP16 mixed precision.
- Submitted independently to **NTIRE 2026 @ CVPR CodaBench international benchmark**; conducted structured ablation studies confirming 1-epoch training (0.793 AUC) outperforms 4-epoch fine-tuning (0.767 AUC) due to out-of-distribution generalization, validated via 10-view TTA ensemble of 3 weighted models.

PyTorch · OpenCLIP · ViT · Kaggle T4 GPU · AdamW

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Statistics, Image Security, Cloud Computing

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & Neuro-Inspired Architectures: LoRA | QLoRA | SFTTrainer | Mamba-Transformer | MoE | vLLM | chain-of-thought reasoning

ML/DL Frameworks & Research Engineering: PyTorch | Hugging Face Transformers | PEFT | TRL | Unsloth | Scikit-learn | OpenCLIP

MLOps, Systems & Data: CUDA debugging | Triton | FP16/BF16 mixed precision | ablation studies | Python | Pandas | Git